

Jerry Zhou

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EDUCATION

University of California, Berkeley | GPA: 4.0/4.0

May 2028

- BS in Electrical Engineering and Computer Science, College of Engineering
- Relevant Coursework: Data Structures and Algorithms, Signals and Information Processing, Physics (E&M, Mechanics), Linear Algebra, Multivariable Calculus

TECHNICAL SKILLS

Software: Python, C/C++, Java, SQL, Scheme, HTML, CSS, JS

ML & Data: PyTorch, JAX/XLA, TensorFlow, LightGBM, NumPy/SciPy, Pandas

Circuit Design/Hardware: PCB Design (Altium, KiCad), Signal Integrity, Power Electronics, Mixed-Signal Design

Tools & Frameworks: Bullet, MuJoCo, CoppeliaSim, Git, ROS2, Fusion 360, Solidworks, MATLAB

EXPERIENCE

Biomimetic Millisystems Lab

Berkeley, CA

Undergraduate Researcher supervised by Dr. Ronald Fearing & Dr. John Atkins

October 2025-Present

- Developed low-level embedded firmware and **Linux-based simulation pipelines** for **ultrasonic motor control on a 7-DOF robotic arm**, enabling real-time closed-loop operation
- Integrated **ROS2 trajectory generation** with physics-based simulation using **PD and impedance controllers with disturbance observers**, improving motion planning accuracy and robustness under load
- Optimized large-scale numerical simulations using **JAX/XLA**, automatic differentiation, and vectorization, **improving computational throughput by 2x** and reducing iteration latency
- Profiled and benchmarked control and simulations workloads to identify memory and performance bottlenecks

Berkeley Haas School of Business

Berkeley, CA

Undergraduate Researcher supervised by Anastassia Fedyk

February 2026-Present

- Built scalable data pipelines to clean, validate, annotate **1M+ records** for downstream analytics and ML models
- Automated consistency checks and deduplication workflows, improving data reliability and reproducibility

Formula Electric at Berkeley

Berkeley, CA

Accumulator Electrical Engineer

September 2025-Present

- Architected a **high-voltage four-layer segment-level BMS PCB** in **Altium** for a **140s3p custom accumulator**, enabling **fault-tolerant** cell voltage and temperature monitoring, cell balancing, and **high-voltage isolation**
- Integrated and debugged the segment board BMS with ADBMS chip, **isoSPI communication**, voltage sensing, and multiplexed temperature monitoring using resistor-divider emulation and **STM32 balancing firmware**.

Innovative Building Energy Control

Lake Forest, CA

Test and Validation Engineer:

December 2025-January 2026

- Commissioned and verified **50+ AC-DC PCBs**, debugging **regulation stability, EMI, and signal integrity**
- Implemented **solar panel telemetry acquisition**, displaying real-time data on LCDs and transmitting via **Wi-Fi**

Magnitude.io

Berkeley, CA

Engineering Consultant: *Magnitude.io*

September 2025-December 2025

- Designed and built a **random positioning machine** to simulate microgravity conditions under a **\$400 BOM**, generating randomized motion profiles using **Fourier-series-based and PRBS trajectories**
- Schematized and brought up a **custom two-layer control PCB** integrating an ESP32, motor drivers, sensors, and display interfaces (I2C, SPI, Bluetooth), with **24V-to-12/5/3.3V power distribution** for operation under load

PROJECTS

Formula One Prediction Model | *Pandas, LightGBM, Python, Weather/Kaggle API*

January 2026

- Constructed a **learning-to-rank model** for F1 race prediction with **gradient-boosted decision trees** achieving **NDCCG@1/3/10** of **0.78/0.77/0.83**, processing **25000+ driver-race combinations** (1950-2020)
- Created **17 temporal features** including exponentially weighted rolling statistics, circuit-specific performance, and teammate comparisons; implemented **shifted rolling windows** preventing data leakage and cold-start outliers

Scheme Interpreter | *Python*

November 2025

- Implemented a full Scheme interpreter in Python, featuring expression parsing, lexical scoping via environment chaining, user-defined procedures, and an extensible primitive system